

PromoGenie – AI-Powered Marketing Copy Generator

Tech Career Accelerator – 2026

Live Application: [PromoGenie](#)

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By

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Introduction

For this project, I developed PromoGenie, an AI-powered marketing copy generator that produces high-quality content for different marketing scenarios. The tool generates product descriptions, social media captions, advertisement text, short promotional messages, and hashtags based on user input.

The purpose of this system is to help users quickly create professional marketing content by selecting different customization options such as tone, audience, platform, and key selling points.

Implementation Architecture

PromoGenie is a full-stack web application built using React for the frontend and Express for the backend. The system follows a structured workflow:

User Input → AI Processing → Output Display → Save to Database

Users enter product details through the interface, and the backend sends this data to the OpenAI GPT model. The AI generates structured marketing outputs, which are displayed to the user and stored in a PostgreSQL database.

The system also includes a history feature that allows users to view previously generated content.

The image shows a web application interface for 'PromoGenie AI'. On the left, there is a sidebar with a 'Generate' button (marked with a star icon) and a 'History' button (marked with a clock icon). The main area is a form for generating marketing copy. It includes a 'Prompt Template' dropdown set to 'Standard — Balanced, all-purpose', a 'Product Name' field with 'Nike T-shirt', and a 'Product Description' field with a sample text about upgrading a wardrobe. Below these are four more dropdowns: 'Target Audience' (Teenagers), 'Key Selling Point' (Stylish Appearance), 'Tone' (Professional), and 'Platform' (Instagram). A large pink 'Generate Copy' button is at the bottom right. A footer box on the left states 'PromoGenie AI Powered by GPT. Generates 5 tailored copy variants instantly.'

Generate

History

Prompt Template

Standard — Balanced, all-purpose

Balanced, all-purpose copy

Product Name

Nike T-shirt

Product Description (min 10 words)

Upgrade your wardrobe with the Nike T-shirt, designed with high-quality fabric for comfort and durability. Perfect for everyday wear, it combines style and reliability.

Target Audience

Teenagers

Key Selling Point

Stylish Appearance

Tone

Professional

Platform

Instagram

Generate Copy

PromoGenie AI
Powered by GPT. Generates 5
tailored copy variants instantly.

Figure 1: Input Form Interface

This screen shows the main user interface of PromoGenie. Users enter product details such as product name and description, and select customization options including target audience, key selling point, tone, platform, and prompt template.

These inputs allow the system to generate tailored marketing content based on user preferences.

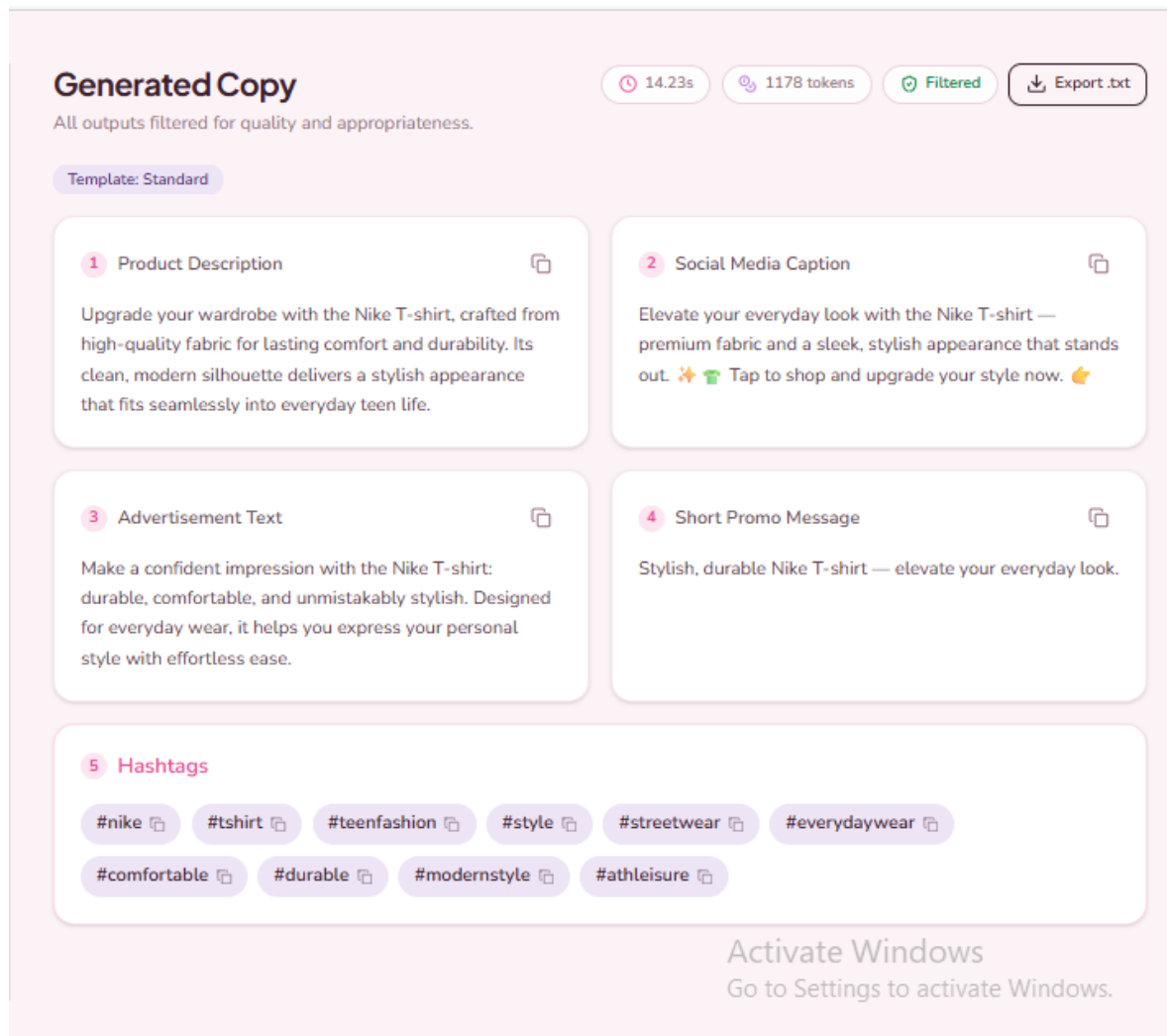


Figure 2: Generated Output Screen

This screen displays the AI-generated marketing content based on the user's input. The system produces multiple outputs, including a product description, social media caption, advertisement text, short promotional message, and relevant hashtags.

These outputs are tailored according to the selected tone, audience, platform, and prompt template

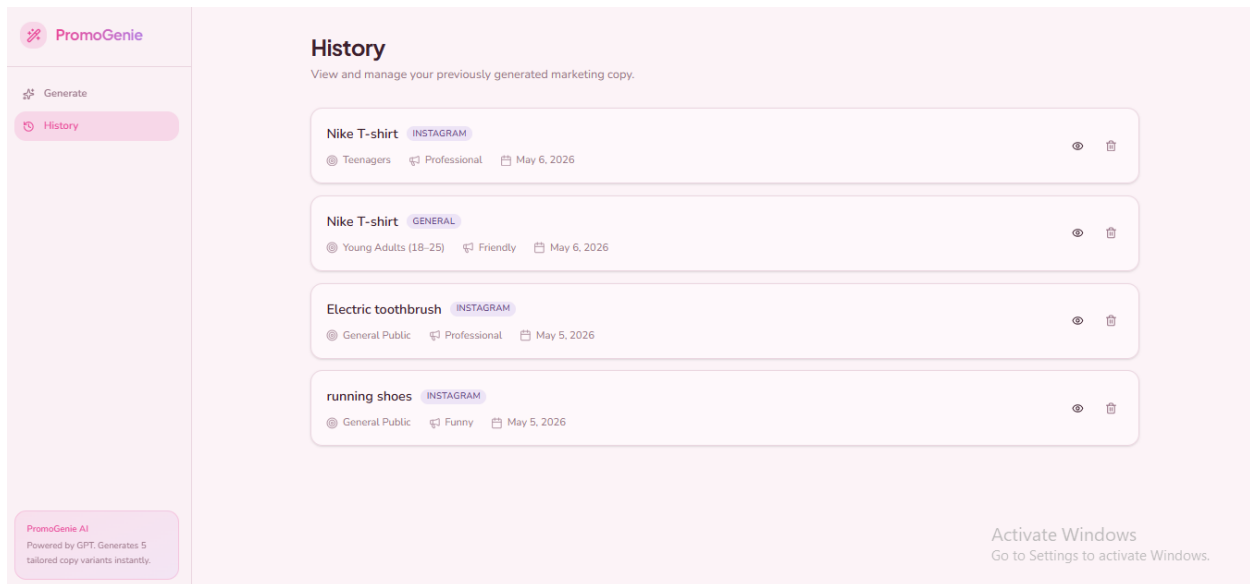


Figure 3: History Page

This screen shows the history of previously generated marketing content. Users can view past results, including product details and generated outputs, allowing them to revisit, compare, and reuse content.

This feature supports result saving and improves usability by keeping a record of all generated campaigns

API Selection Rationale

The OpenAI GPT model (gpt-4.1-mini) was selected because it provides accurate and high-quality text generation.

The main reasons for choosing this API include:

- Strong instruction-following ability
- Ability to generate structured outputs
- Token usage tracking for performance monitoring
- Built-in content safety features

These features make it suitable for generating reliable and professional marketing content.

Prompt Engineering Methodology

The system uses a structured prompt engineering approach to ensure consistent and high-quality outputs.

Each prompt includes:

- A role definition where the AI acts as a marketing expert
- Instructions based on tone, audience, and platform
- A structured output format (5 content types)
- Safety rules to prevent inappropriate content

This ensures that all outputs are relevant, clear, and aligned with user inputs.

Prompt Templates

The system includes five optimized prompt templates:

1. Standard Template

Used for general-purpose marketing content with a balanced tone.

2. Storytelling Template

Focuses on emotional and narrative-driven content to engage the audience.

3. Feature-Focused Template

Highlights product specifications and benefits in a clear and direct way.

4. Urgency & Sale Template

Creates persuasive content using urgency and limited time offers.

5. Minimalist Template

Generates short and impactful content using minimal words.

Sample Inputs

The following example shows how users interact with the system:

- Product Name: Nike T-shirt
- Product Description: A high-quality t-shirt designed for comfort and everyday style for young adults

- Target Audience: Young Adults (18–25)
- Key Selling Point: High Quality
- Tone: Friendly
- Platform: Instagram
- Prompt Template: Standard

Sample Outputs

Below are five examples generated using different templates:

Sample 1 – Standard Template

Product Description:

Upgrade your wardrobe with the Nike T-shirt, designed with high-quality fabric for comfort and durability. Perfect for everyday wear, it combines style and reliability.

Social Media Caption:

Stay stylish every day, Grab your Nike T-shirt now!

Advertisement Text:

Experience comfort and style with the Nike T-shirt. Built for everyday confidence.

Short Promo Message:

Style meets comfort.

Hashtags:

#Nike #TShirt #Streetwear #Fashion #EverydayStyle

Sample 2 – Storytelling Template

Product Description:

Every journey begins with a choice. The Nike T-shirt is designed for those who want comfort and confidence in every step of their day.

Social Media Caption:

Your story starts with style

Advertisement Text:

Feel confident every day with a t-shirt that matches your lifestyle.

Short Promo Message:

Wear your story.

Hashtags:

#Lifestyle #Fashion #Comfort #EverydayWear

Sample 3 – Feature-Focused Template

Product Description:

The Nike T-shirt features high-quality fabric, durable stitching, and a comfortable fit designed for everyday use.

Social Media Caption:

Quality you can feel. Style you can trust.

Advertisement Text:

Designed with premium materials for long-lasting comfort and performance.

Short Promo Message:

Built for quality.

Hashtags:

#HighQuality #Nike #Clothing #Durable

Sample 4 – Urgency & Sale Template

Product Description:

Get your Nike T-shirt now while stocks last. Limited availability for a high-quality everyday essential.

Social Media Caption:

Limited stock! Don't miss out!

Advertisement Text:

Act fast and grab your Nike T-shirt before it's gone.

Short Promo Message:

Hurry - limited time only!

Hashtags:

#LimitedOffer #Sale #ActNow #Nike

Sample 5 – Minimalist Template

Product Description:

Nike T-shirt. Simple. Comfortable. Reliable.

Social Media Caption:
Clean style. Everyday comfort.

Advertisement Text:
Less words. More quality.

Short Promo Message:
Wear better.

Hashtags:
#MinimalStyle #Nike #CleanFashion

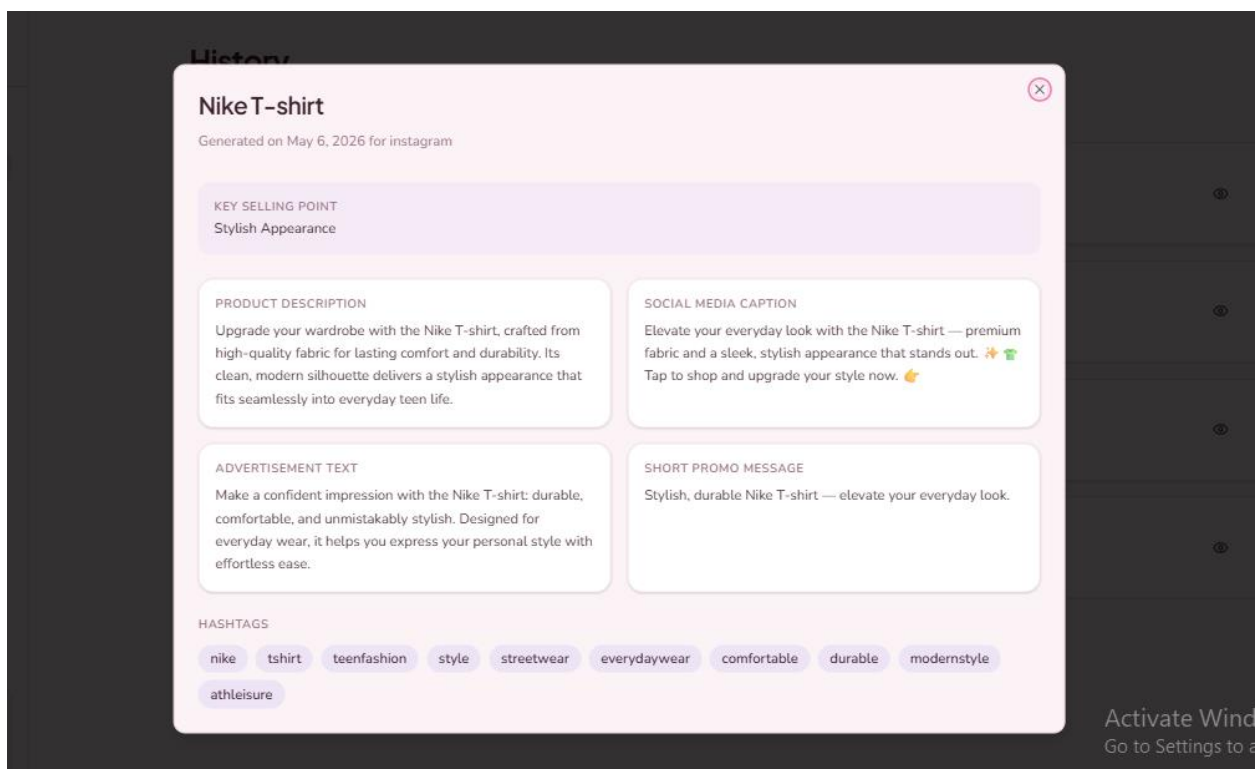


Figure 4: Detailed Output Example

This screen provides a detailed view of a single generated result. It shows all five content types clearly, including product description, caption, advertisement text, promotional message, and hashtags.

This demonstrates the system's ability to generate complete and structured marketing content in one request.

Input Validation and Output Filtering

The system includes input validation to ensure accurate results:

- All fields are required
- Product description must contain at least 10 words
- Dropdown menus ensure consistent inputs

Output filtering ensures:

- Content is professional and appropriate
- Tone matches user selection
- No harmful or irrelevant content is generated

Performance Optimization Techniques

The system tracks performance using:

- Generation time (2–5 seconds)
- Token usage (for cost monitoring)

Additional optimizations include:

- Input validation to reduce unnecessary API calls
- Efficient data handling and storage
- Structured outputs for faster processing

Limitation Management Strategies

Some limitations include:

- API rate limits
- Variation in output quality based on input
- Response time delays

These are managed through:

- Error handling messages
- Minimum input requirements
- Loading indicators

Comparison of Prompt Templates

Each template produces different results:

- Storytelling → emotional and engaging
- Feature-focused → detailed and technical

- Urgency → persuasive and action-driven
- Minimalist → short and impactful
- Standard → balanced and versatile

This shows how prompt design affects output quality.

Comparison Matrix of Prompt Templates

Template	Strength	Best Use Case	Output Style
Standard	Balanced	General marketing	Neutral
Storytelling	Emotional	Lifestyle brands	Narrative
Feature-Focused	Technical	Tech products	Detailed
Urgency & Sale	High conversion	Promotions	Persuasive
Minimalist	Premium feel	Luxury brands	Short & clean

Conclusion

PromoGenie successfully meets all project requirements by providing a functional AI-powered marketing copy generator. The system demonstrates effective prompt engineering, structured design, and practical use of generative AI.